YAKEEN

NEET 2.0 2027

Physics | Lecture - 6

Basic Maths and Calculus

By- Manish Raj (MR Sir)



Todays Goal

straight line, Parabola, hyperbola, circle & ellipse.



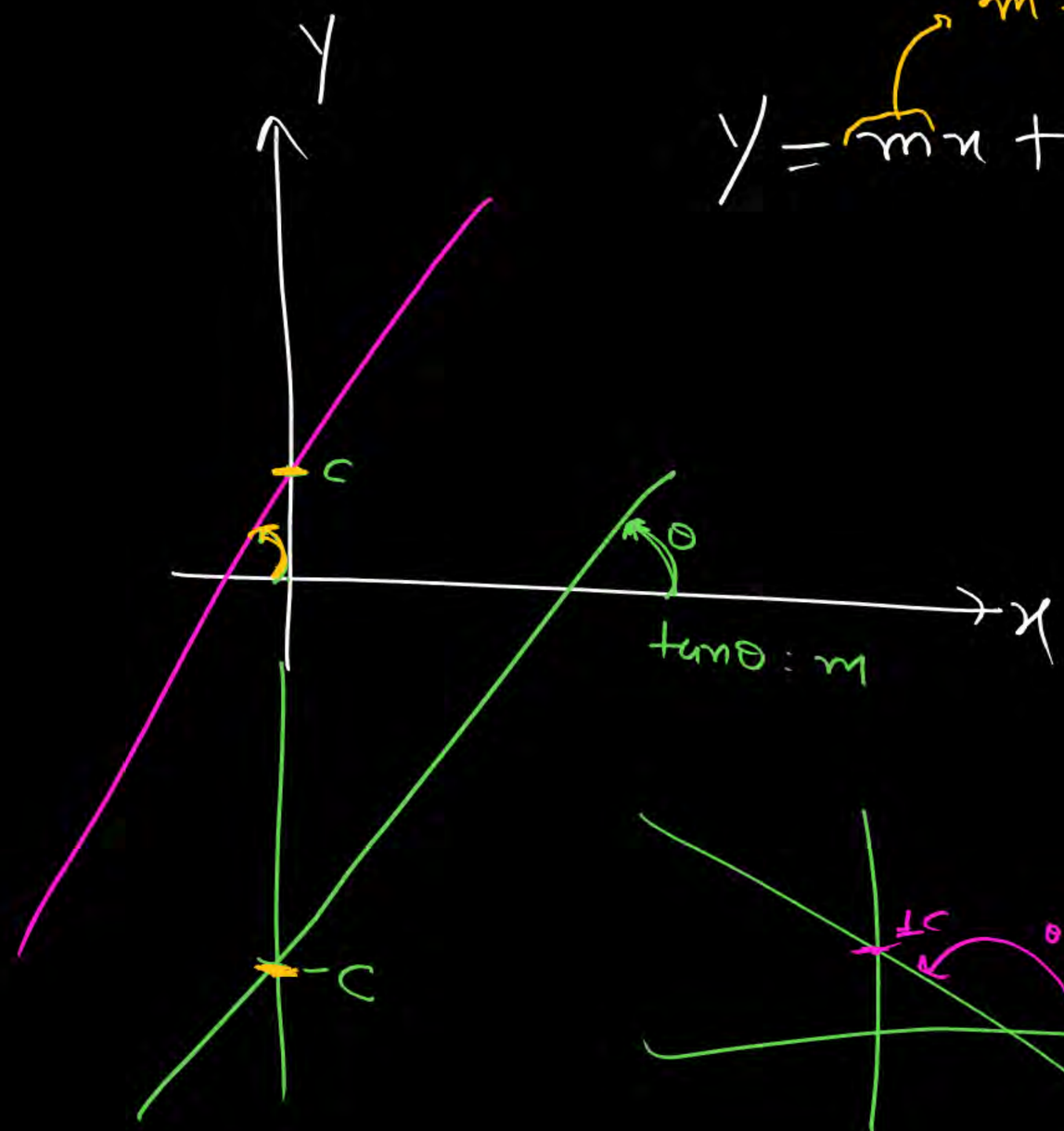
Sharsh Assignment 1 & 2

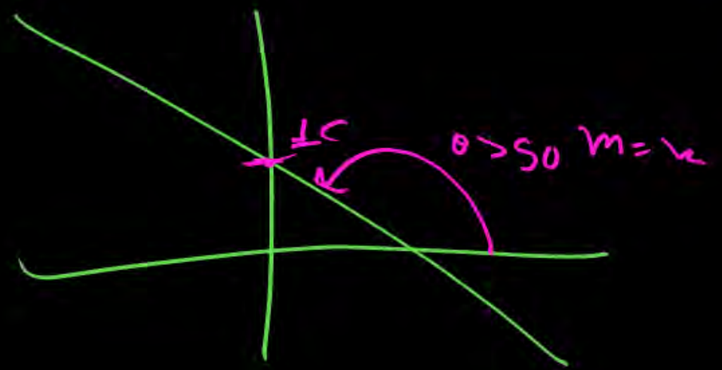
Join

$m = \underline{\text{slope}} = \tan\theta = \frac{\Delta y}{\Delta x}$

$y = mx + c$

c → y-Intercept





$\tan\theta = m = \frac{y}{x}$

$\tan\theta = \frac{\Delta y}{\Delta x} = \frac{y-0}{x-0}$

Copy

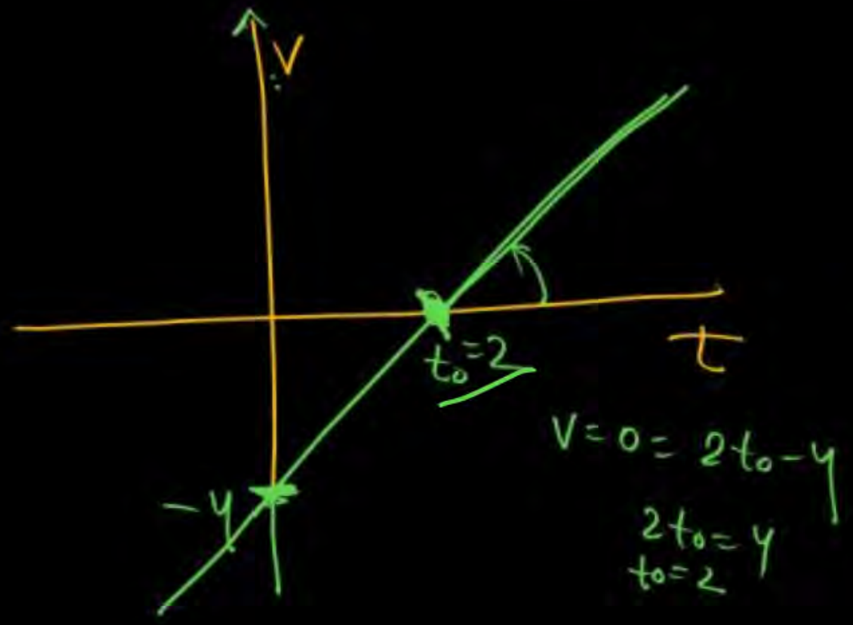
Velocity of object $V = 2t - 4$ then Draw Velocity time graph

$V = 2t - 4$

$y = mx + c$

$c = -4$

$m = 2 = \tan\theta$



$V = 0 = 2t_0 - 4$

$2t_0 = 4$

$t_0 = 2$

$V = 2t - 4$

$y = mx + c$

$V = 2t - 4$

$V = 0 = 2t - 4$

$2t = 4$

$t = 2$

$t = 0$

$V = -4$

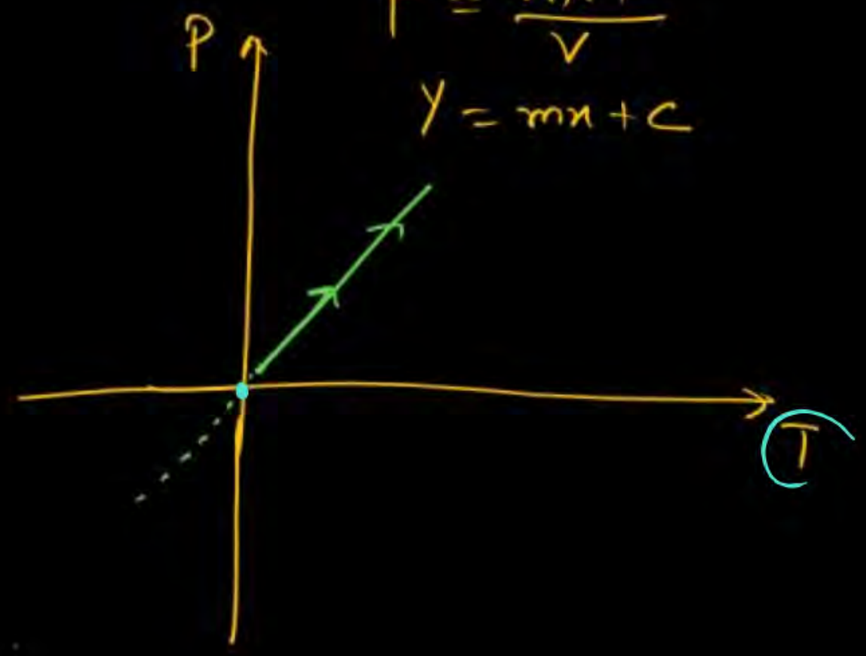


Copy gf $PV = nRT$ then draw "P and T" if V=con^n

$(P)V = nR(T)$

$P = \frac{nRT}{V}$

$y = mx + c$



$PV = nRT$

$P = \frac{nR}{V}T$

$y = mx + C$

$m = \frac{nR}{V}$ = tanθ

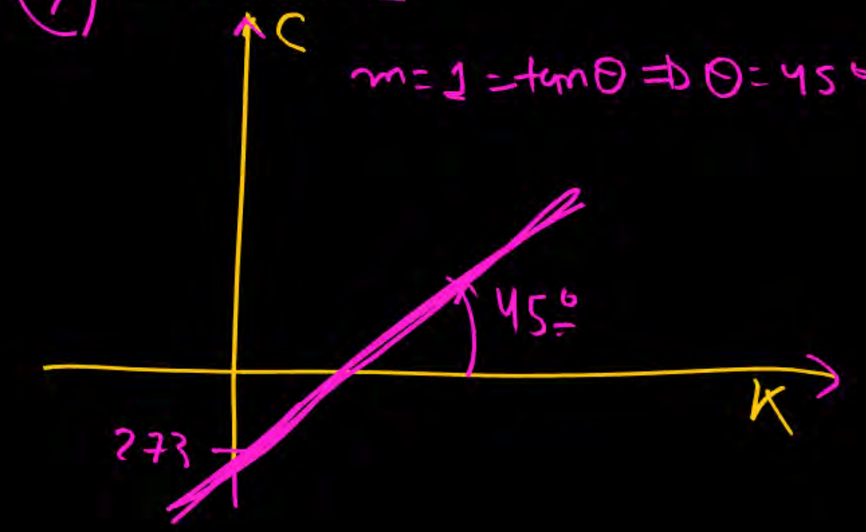
If equation $\frac{C-0}{100-0} = \frac{K-273}{373-273}$ then draw graph b/w C & K

JEE main

$$\frac{C}{100} = \frac{K-273}{100}$$

$C = K - 273$

$y = mx + c$



Question

Find equation of the line which makes intercept +4 and 5 on the x and y-axis.

1. $5x + 4y + 20 = 0$
2. $4x + 5y - 20 = 0$
3. $4y - 5x = -20$
4. $4x + 5y + 20 = 0$

No copy

$y = 5 \quad (x = 0)$

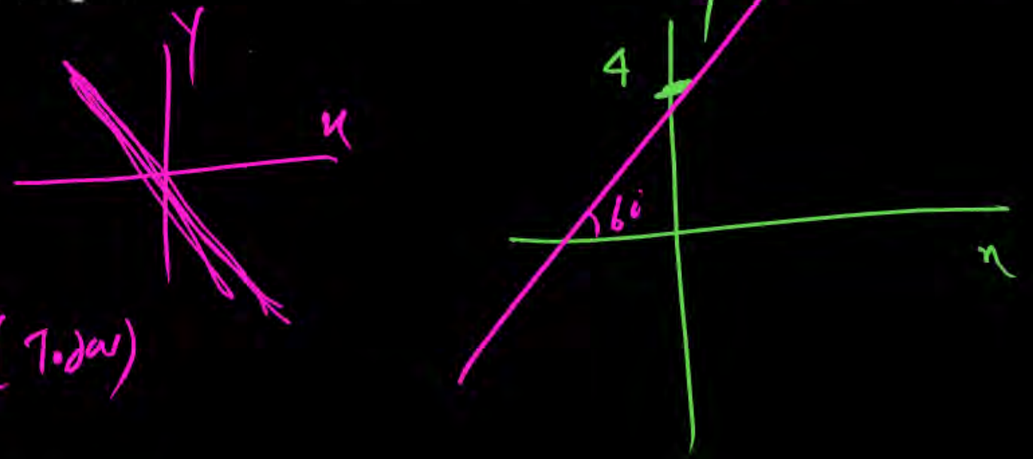
$x = 4 \quad (y = 0)$

$5x + 4y - 20 = 0$ correct eqn

Question

Draw graph between y and x for given equation:

(i) $y = \sqrt{3}x + 4$



(ii) $y = -x$

(iii) $y = |x|$ (Today)

(iv) $\frac{x}{3} + \frac{y}{4} + 1 = 0$

$\frac{x}{3} + \frac{y}{4} = -1$

$\frac{y}{4} = -\frac{x}{3} - 1$

$y = -\frac{4}{3}x - 4$

copy

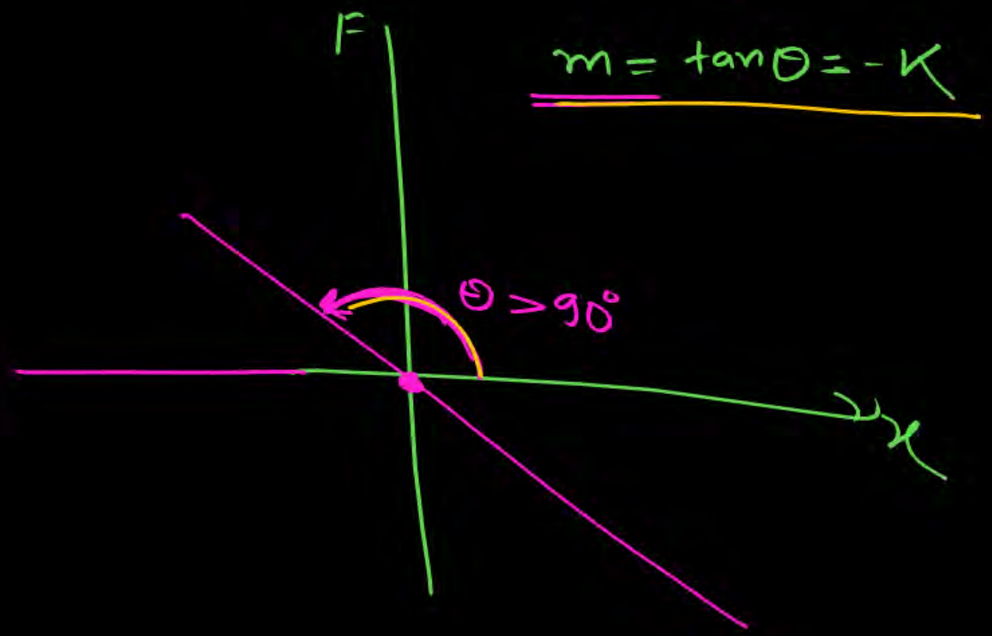
(a) Spring force $\boxed{F = -Kx}$ Draw graph b/w 'F' and 'x'.

$F = -Kx$

$y = mx + c$

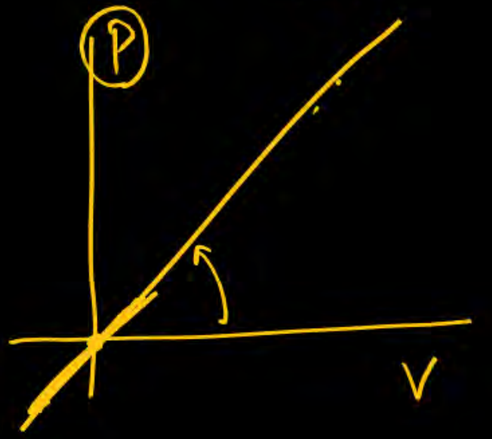
$c = 0$

$m = \tan\theta = -K$



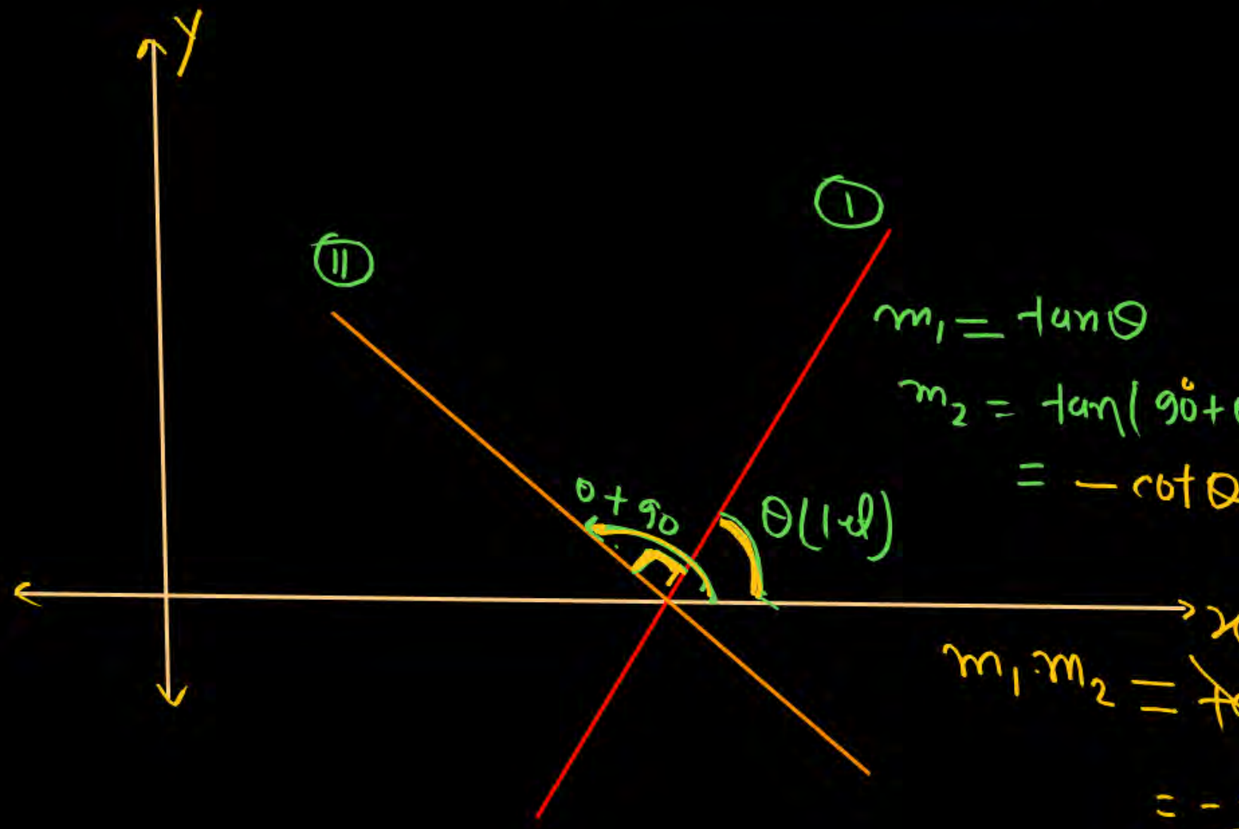
$y = mx + c$

$P = mV$



Q. If Two straight line **Perpendicular** to each other then prove that product of their slope is "-1".

$$m_1 \cdot m_2 = -1$$



$m_1 = \tan\theta$

$m_2 = \tan(90^\circ + \theta)$

$= -\cot\theta$

$m_1 \cdot m_2 = \tan\theta(-\cot\theta)$

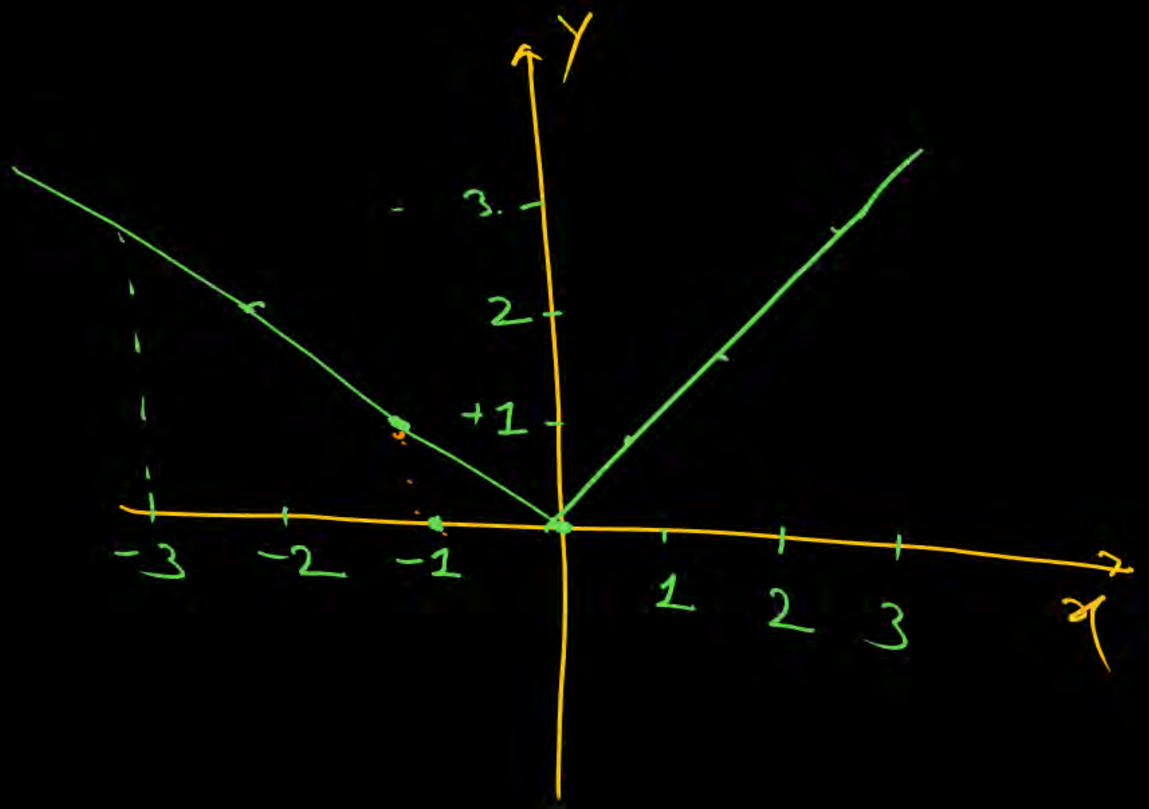
$= -1$

$\tan\theta = \frac{1}{\cot\theta}$

$y = |x|$ ⟹ value of y must be +ve

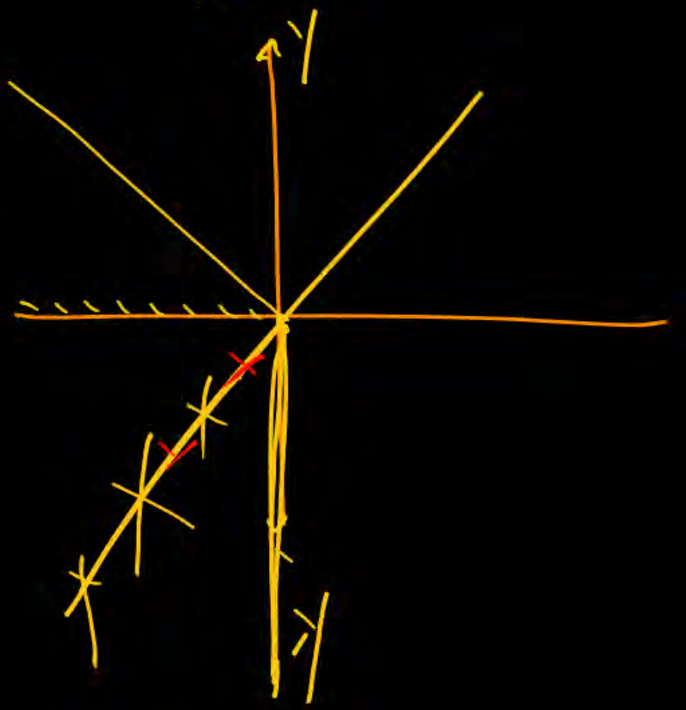
Draw graph b/w 'y's 'x'

$y = |x|$ → $y = x$

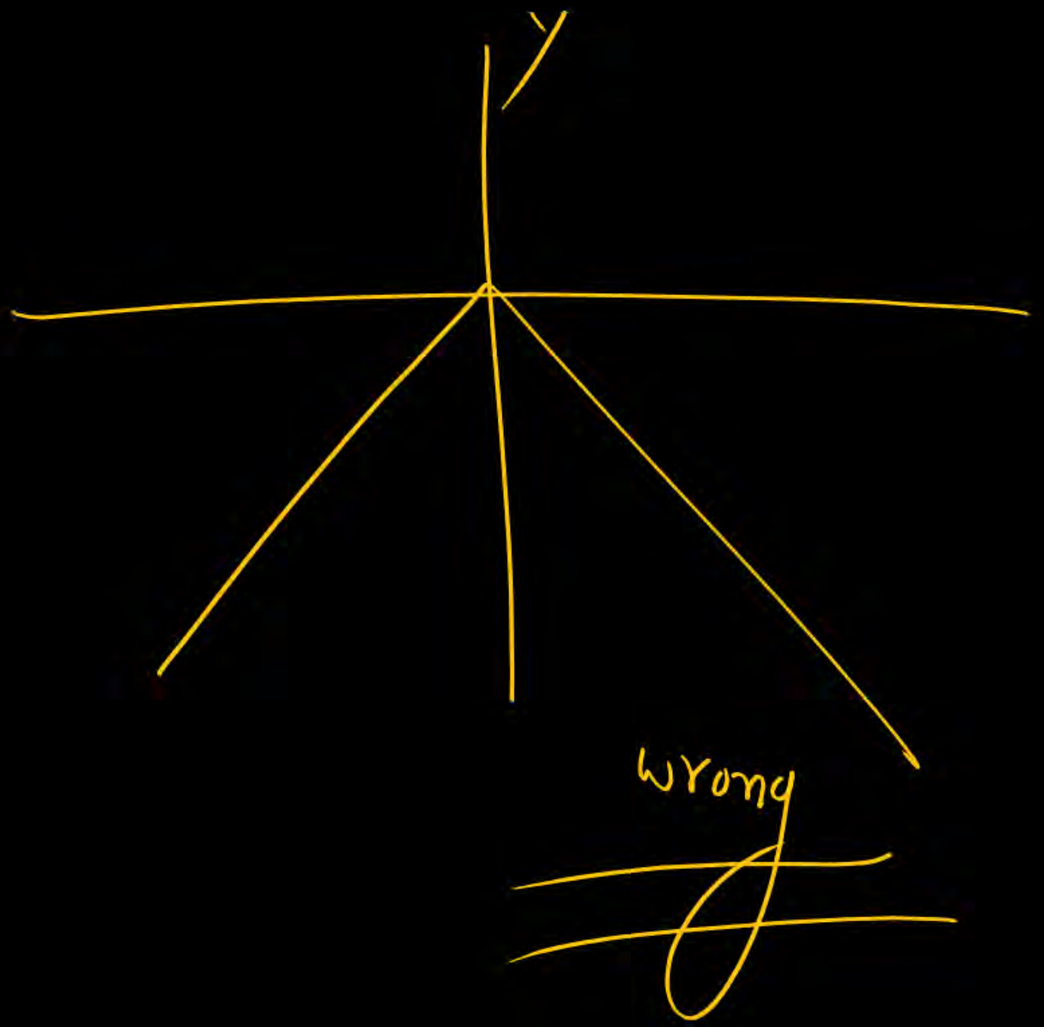


y	x
(0	0)
1	1
2	2
3	3
1	-1
2	-2
3	-3

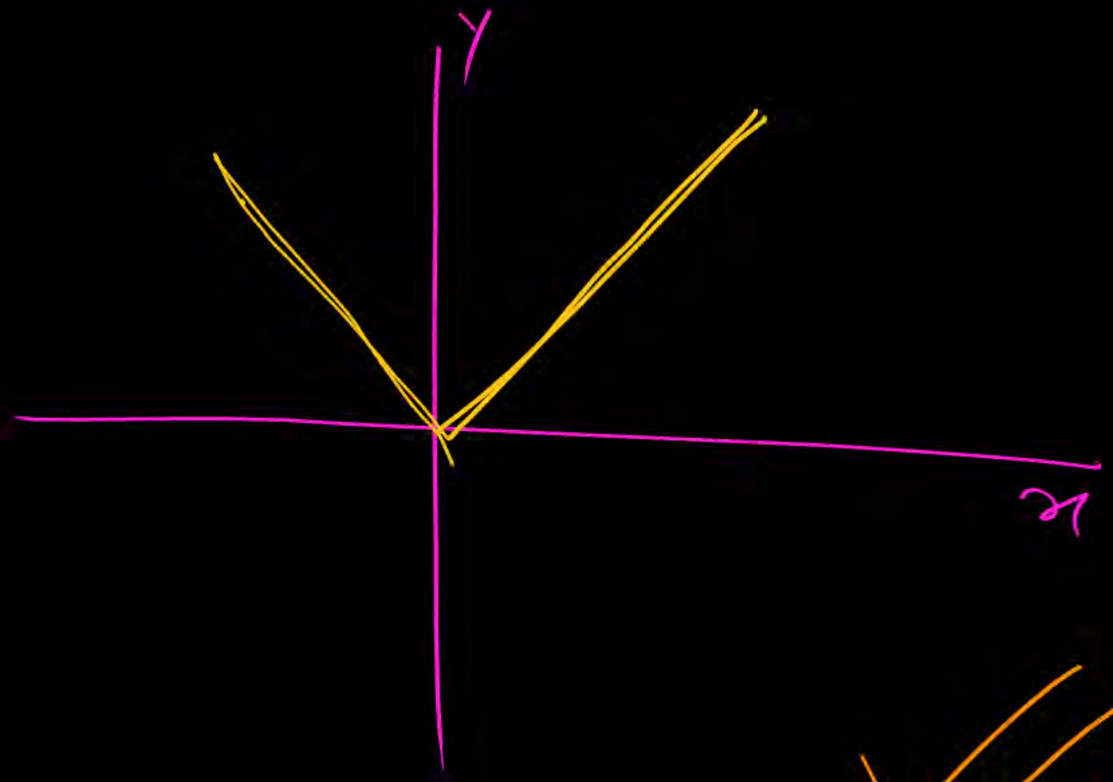
MR*



$y = |-x|$ Draw graph B/w y & x.

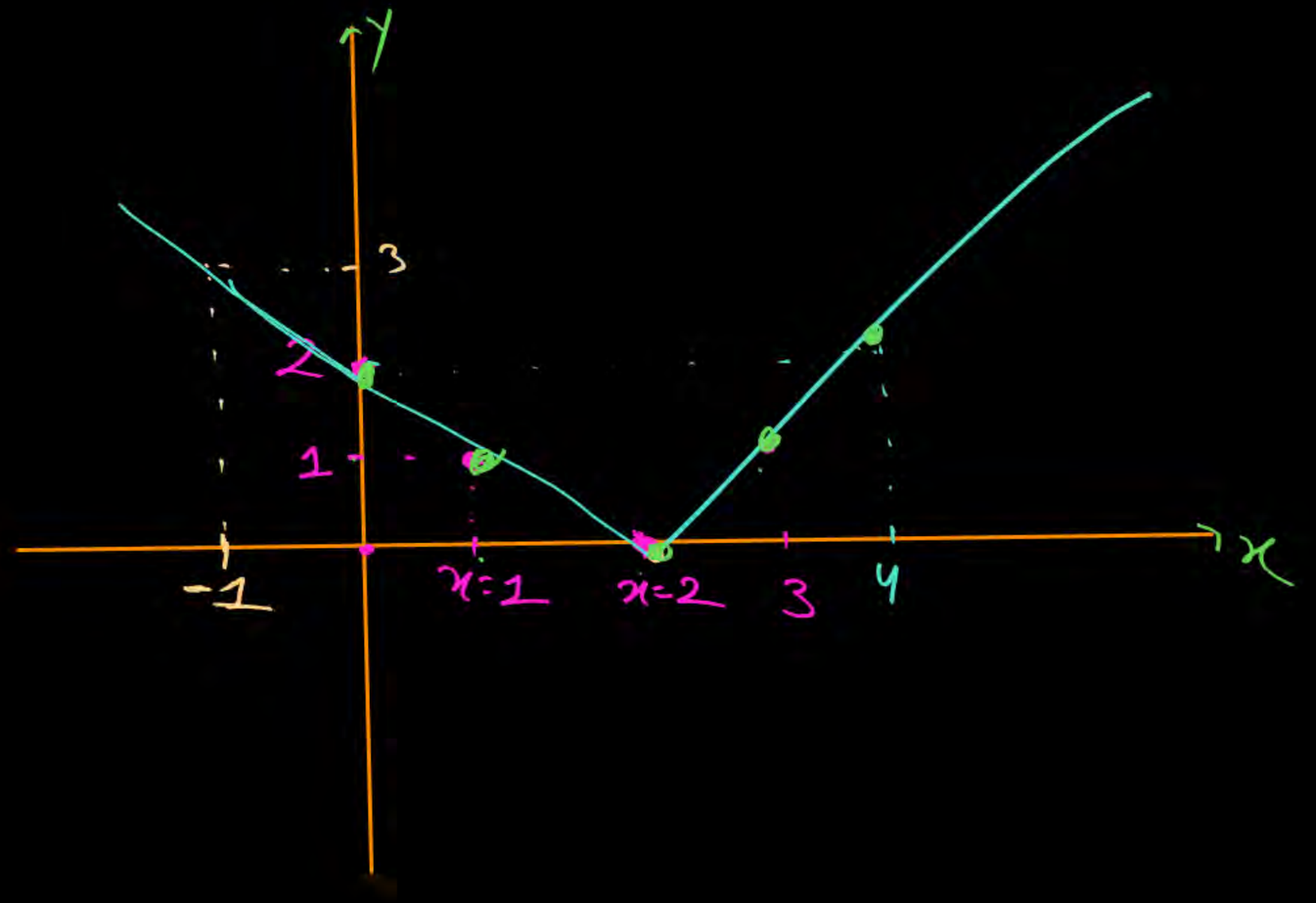


Wrong



$$y = |x-2|$$

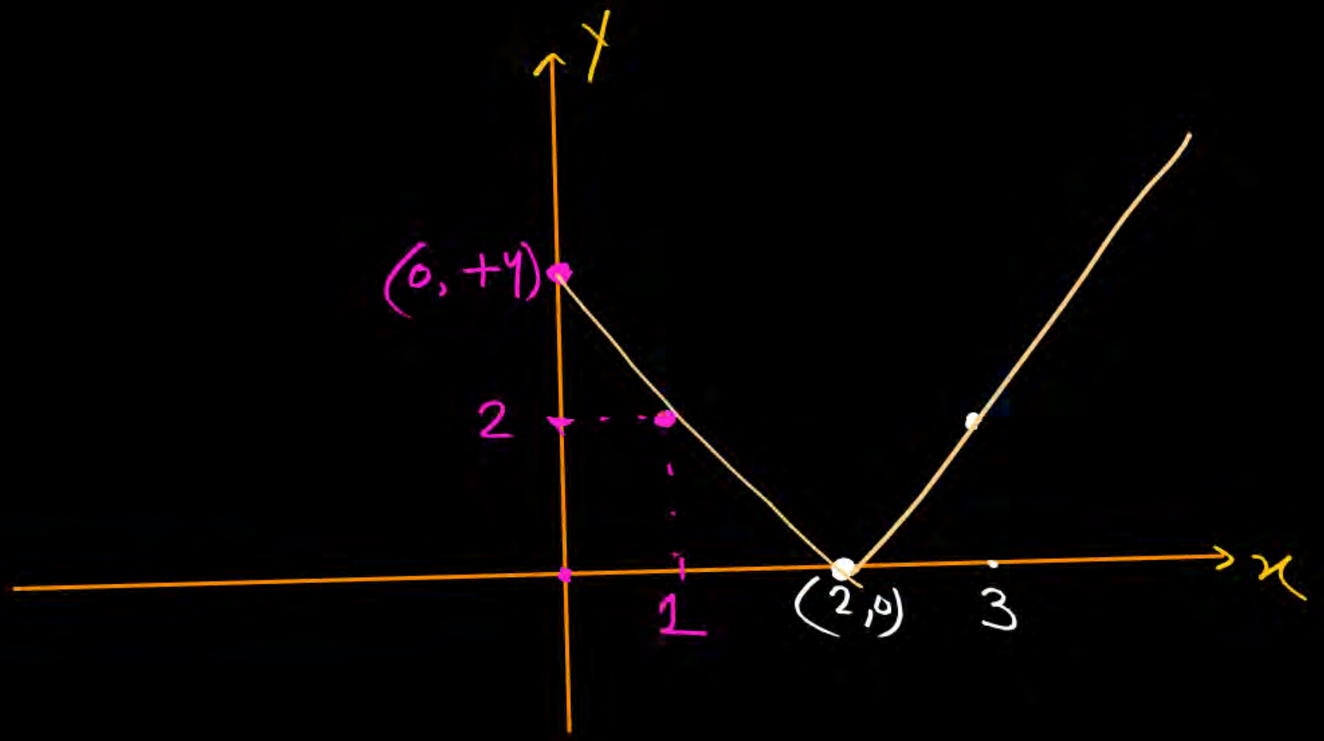
y	x
2	0
1	1
0	2
1	3
2	4



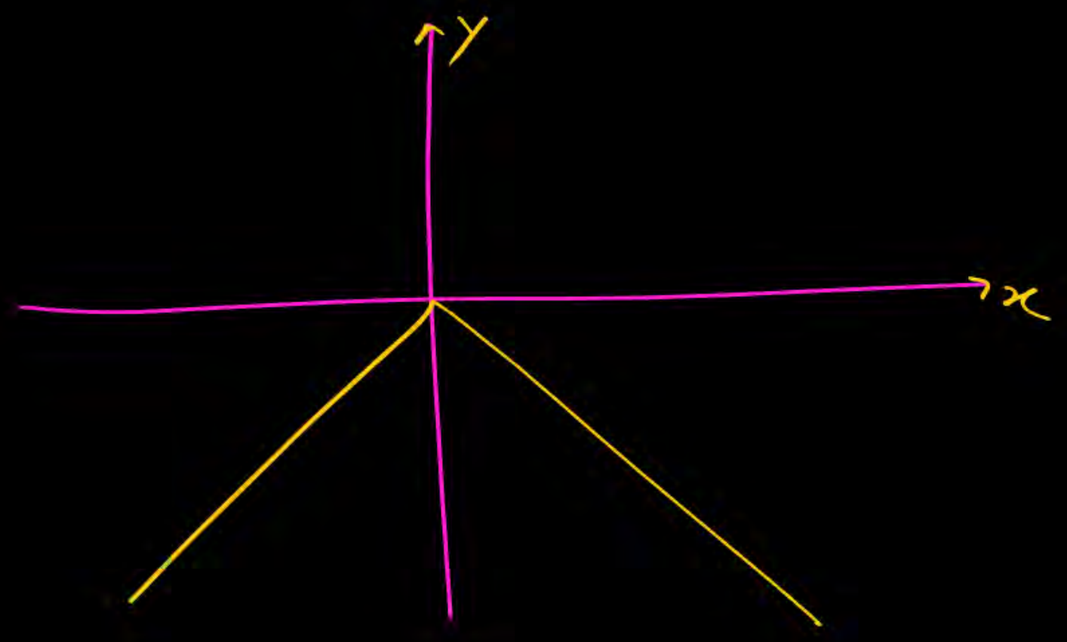
$$y = |4 - 2x|$$

Put $x = 0 \rightarrow y = 4$

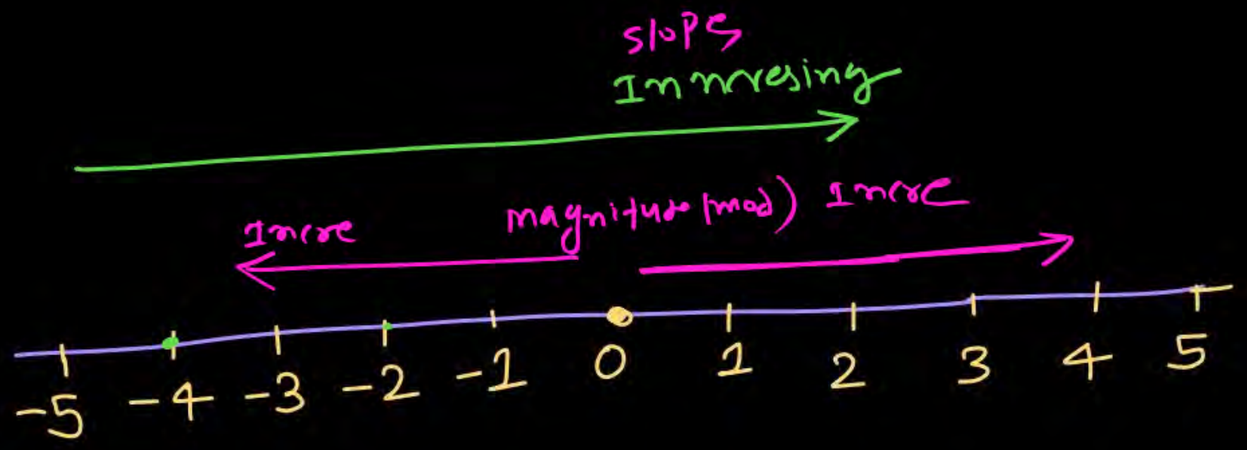
$x = 1$	$y = 2$
$x = 2$	$y = 0$
$x = 3$	$y = 2$



$$y = -|x|$$



Variation of slope and magnitude of slope :-



slope

$-5 < 5$

magnitude

$|-2| < |-4|$

Magnitude (mod) of slope.

$|-5| = |5|$

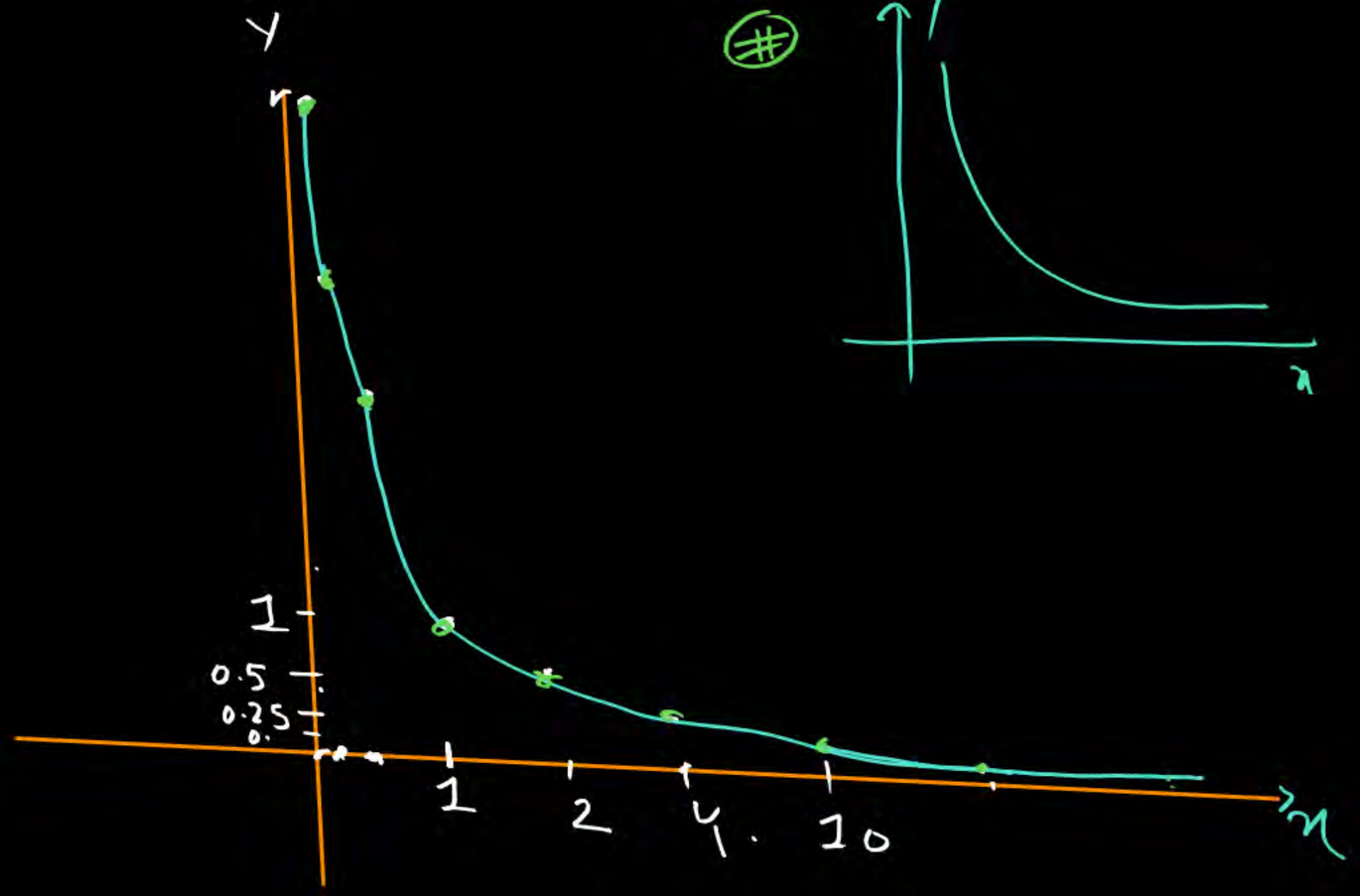
slope

$-2 > -4$

Rectangular hyperbola :- [Inversely Proportional relation]

(↓) $y = \frac{1}{x}$ (1)

y	x
1000	0.001
100	0.01
10	0.1
1	1
0.5	2
0.25	4
0.1	10
0.01	100



copy at home

MR* Box for graph

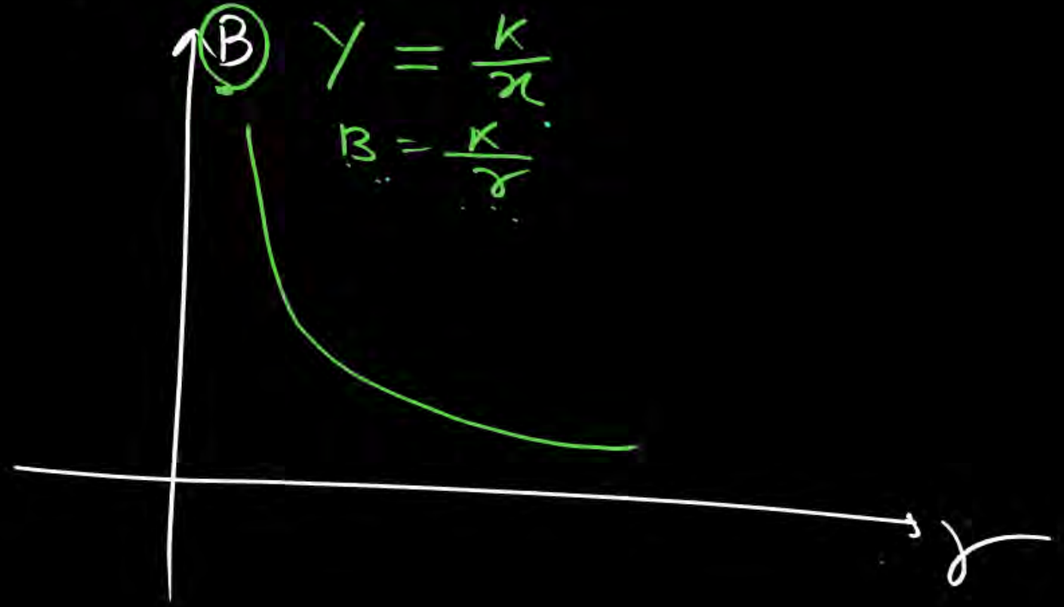
- Kisi bhi equation se graph Draw karne ke liye x ke different value Put kar ke y ki value find karo.
- then Locate Co-ordinate of (x y) in graph as a point
- connect all Point You will get graph

Q12

$B = \frac{\mu_0 I}{2\pi r} = \frac{K}{r}$ ← constant (K)

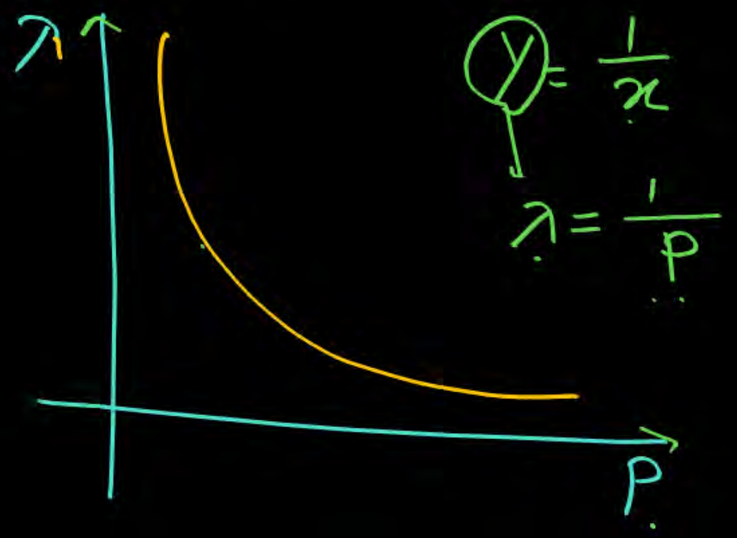
Magnetic field due to infinite wire :-

Draw B/r graph



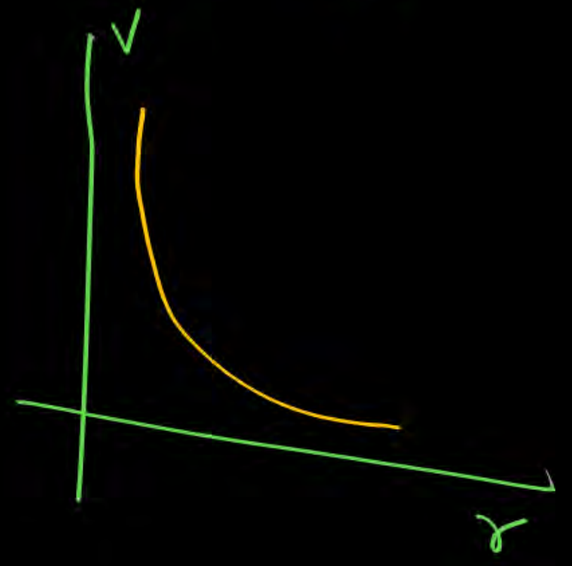
Draw graph b/w De-broglieee wavelength & momentum:-

$$\lambda = \frac{h}{p}$$ (rel^n.)

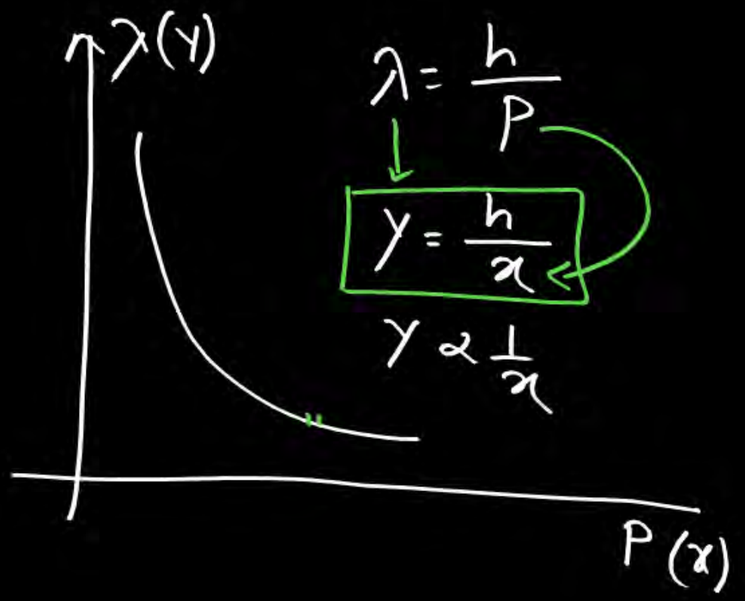


Electric Potential

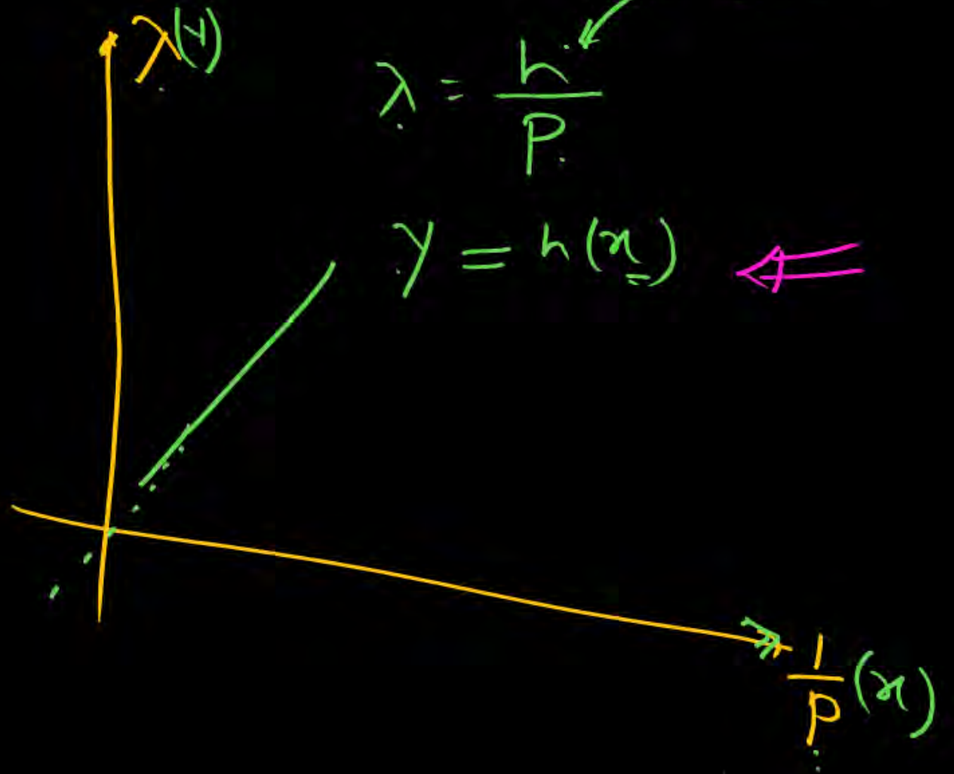
$V = Kq/r$



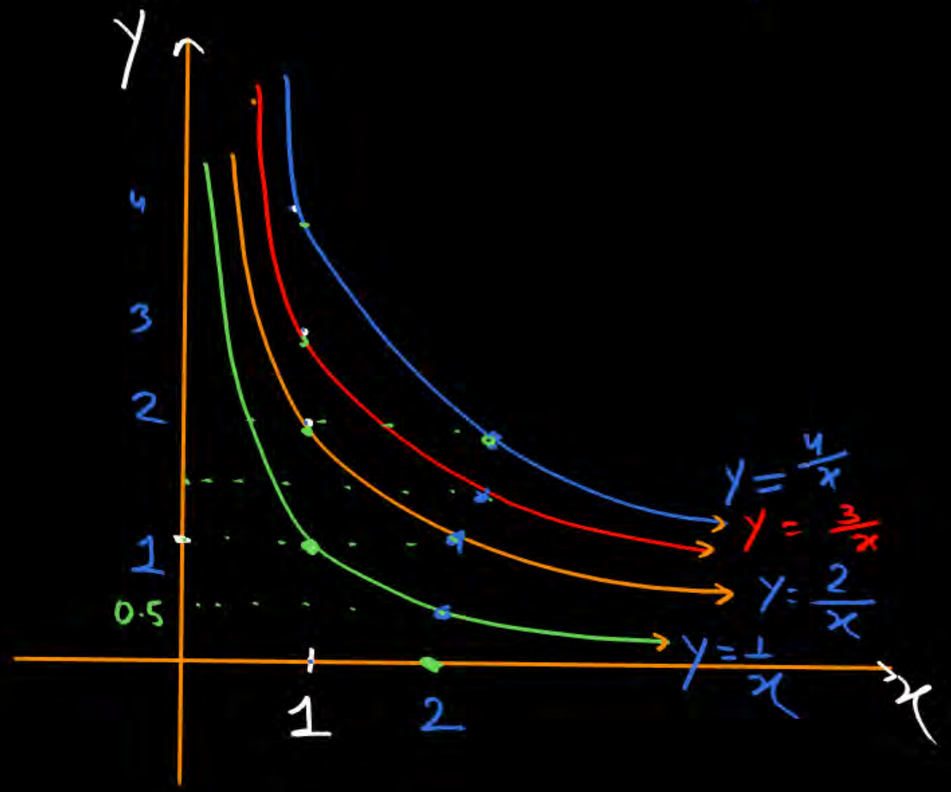
Draw graph b/w "λ" and "$\left(\frac{1}{P}\right)$" from equation $\left[\lambda = \frac{h}{P}\right]$ given in question



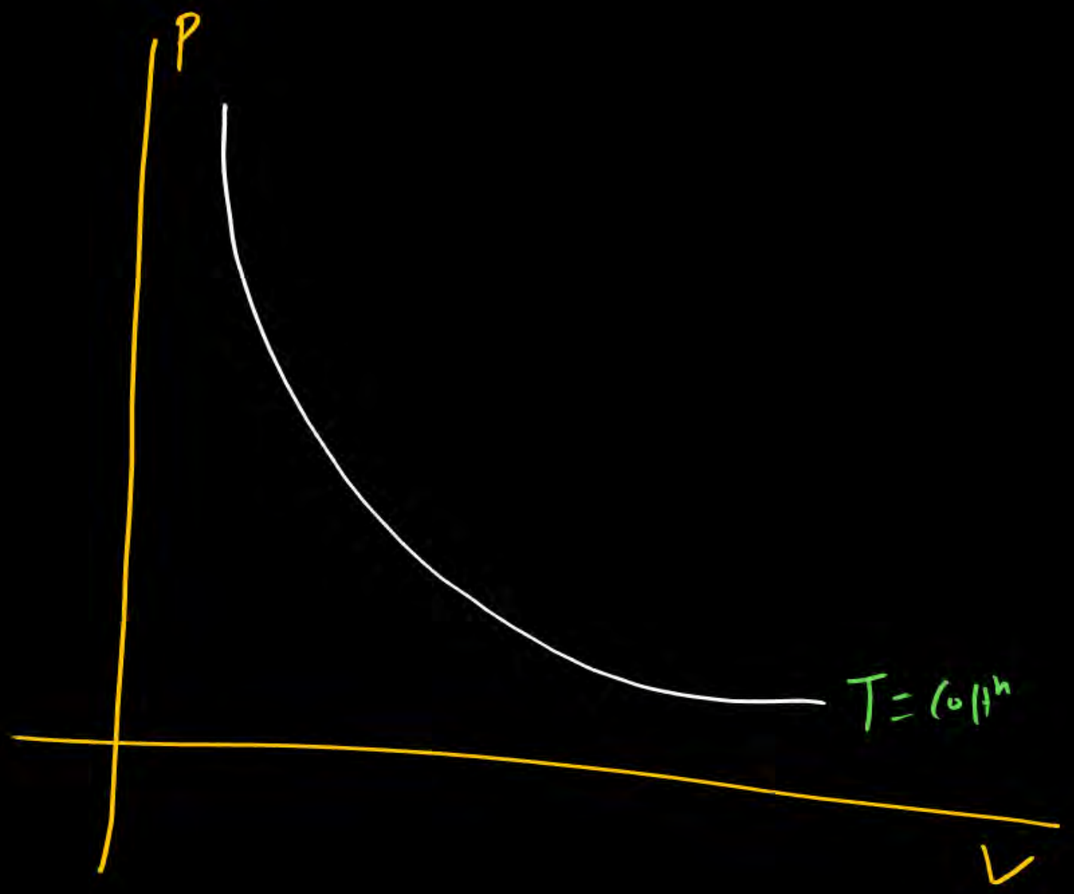
Plancks eqn



① $y = \frac{1}{x}$ ② $y = \frac{2}{x}$ ③ $y = \frac{3}{x}$ ④ $y = \frac{4}{x}$ } all are Rectangular hyperbola



→ here No two graph will cut - each other



Draw PV graph for isothermal (T = const) process :—

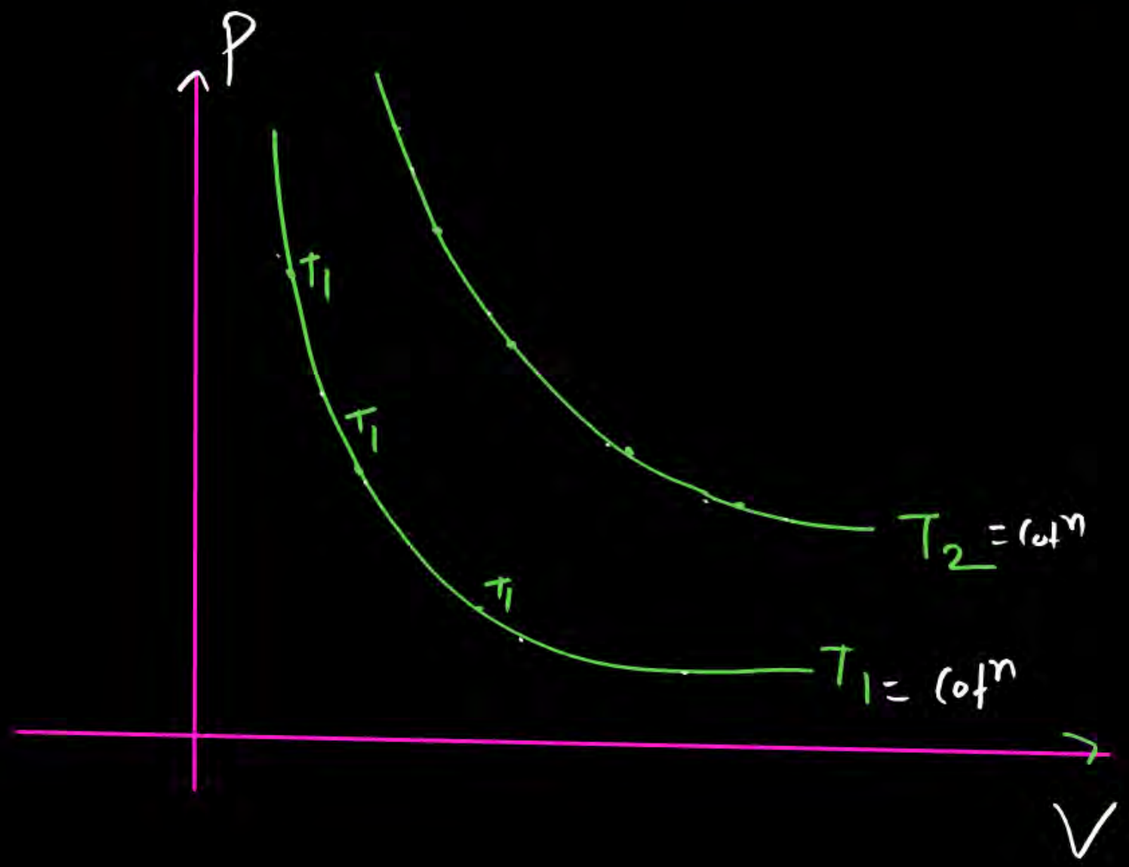
$$PV = nRT$$

$$P = \frac{[nRT]}{V} \propto \frac{K(\text{const})}{V}$$

$$\left[P \propto \frac{1}{V}\right]$$

$$y \propto \frac{1}{x}$$

(8) Two Isothermal (P-V) curve is given at Tempr T_1 & T_2 as shown. then which is at higher Tempr



$$"P" = \frac{(nRT)}{"V"}$$

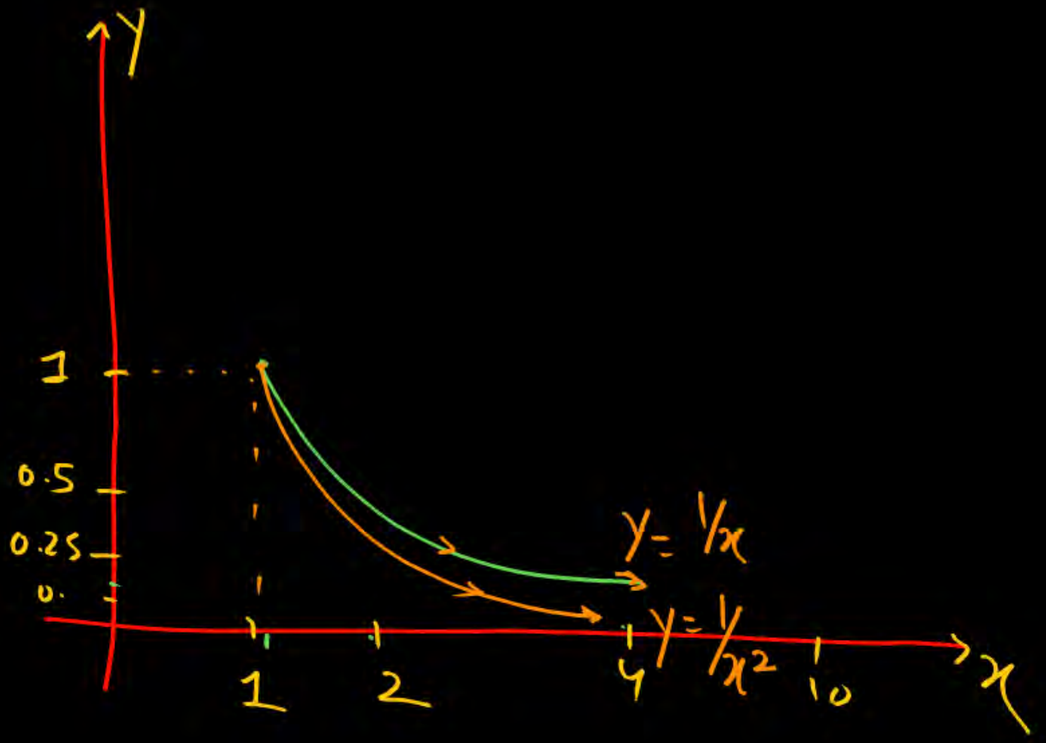
$(T_2 > T_1)$ ⟸

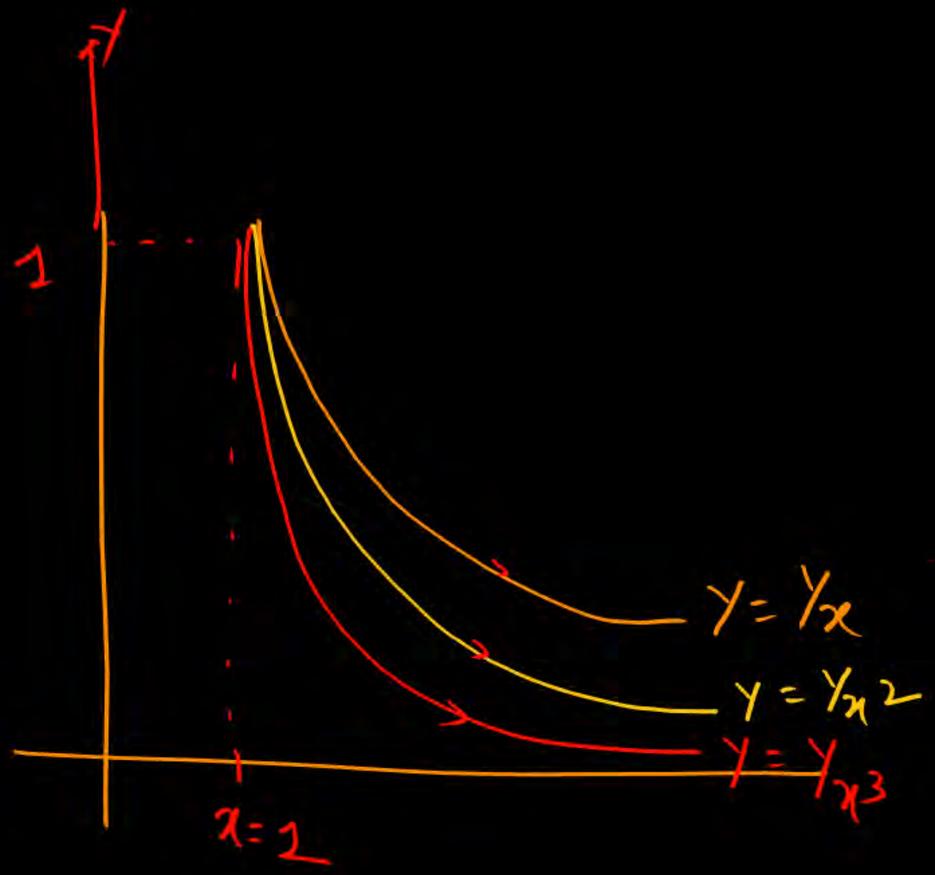
Answer

$y_1 = \frac{1}{x}$ (Rectangular hyperbola)

$y_2 = \frac{1}{x^2}$ (like Rectangular hyperpola)

y_1	y_2	x
1	1	1
0.5	0.25	2
0.25	0.0625	4
0.1	0.01	10



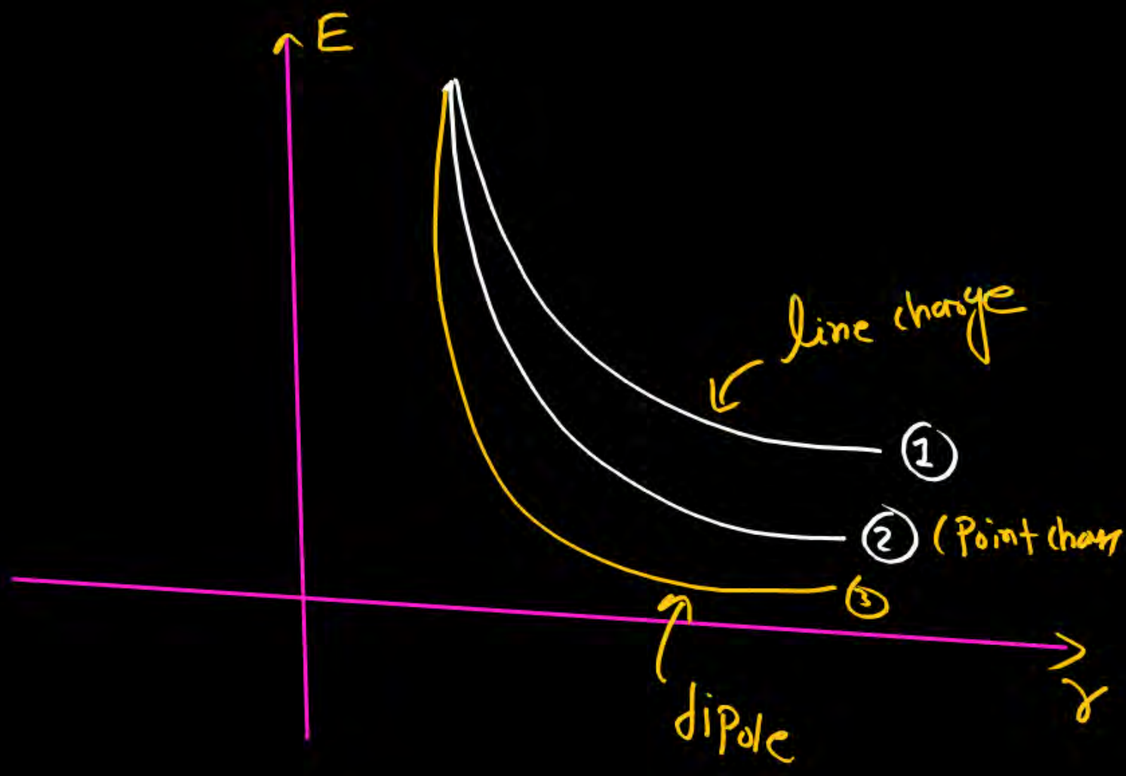

y
1
y = 1/x
y = 1/x^2
y = 1/x^3
x = 1

(0) Electric field due to

(1) linear charge $E_1 = \frac{2K\lambda}{r} \propto \frac{1}{r}$

(2) Point charge $E_2 = \frac{Kq}{r^2} \propto \frac{1}{r^2}$

(3) due to dipole $E_3 = \frac{Kq}{r3} \propto \frac{1}{r^3}$

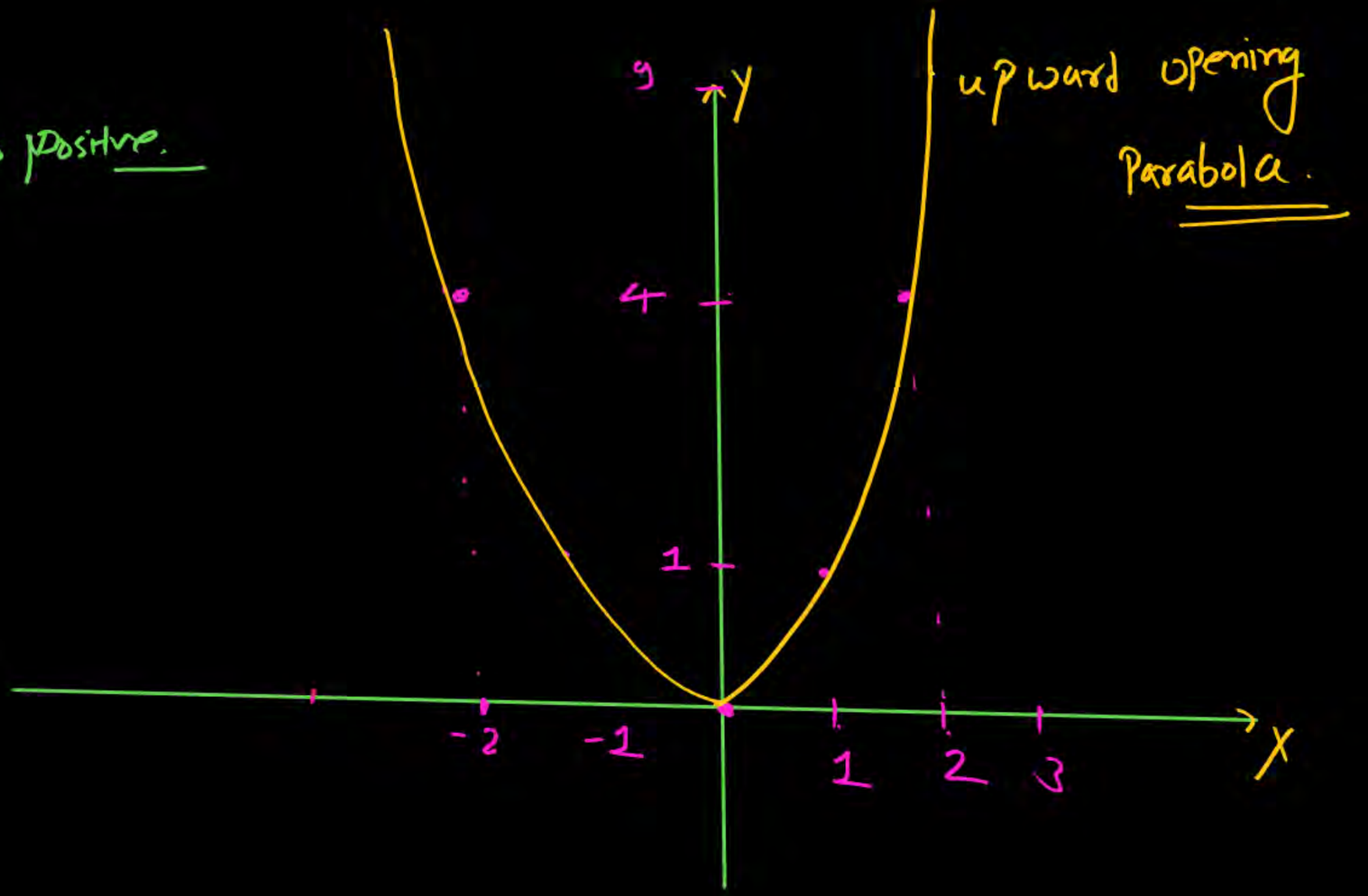


Parabolic graph :-

$$y = x^2$$

→ y → always positive.

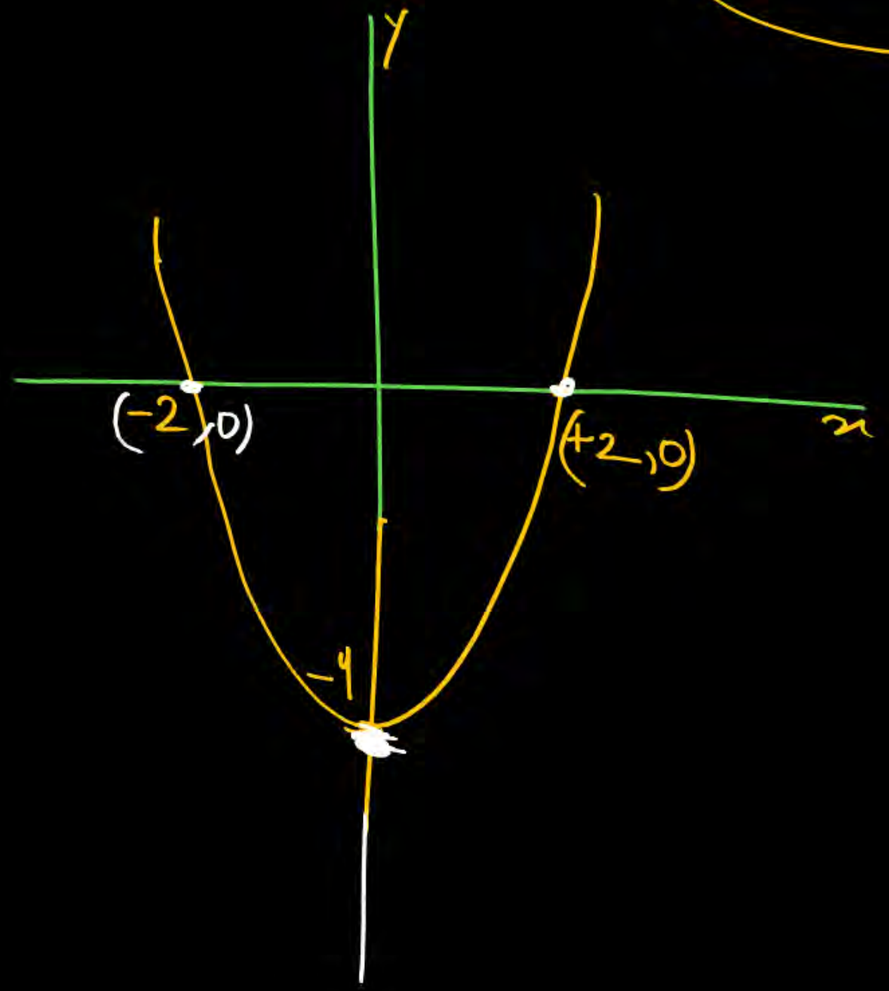
y	x
0	0
1	1
4	2
9	3
1	(-1)
4	-2
9	-3



upward opening Parabola.

$$y = x^2 - 4$$

↳ upward Parabol.



Put $y = 0$

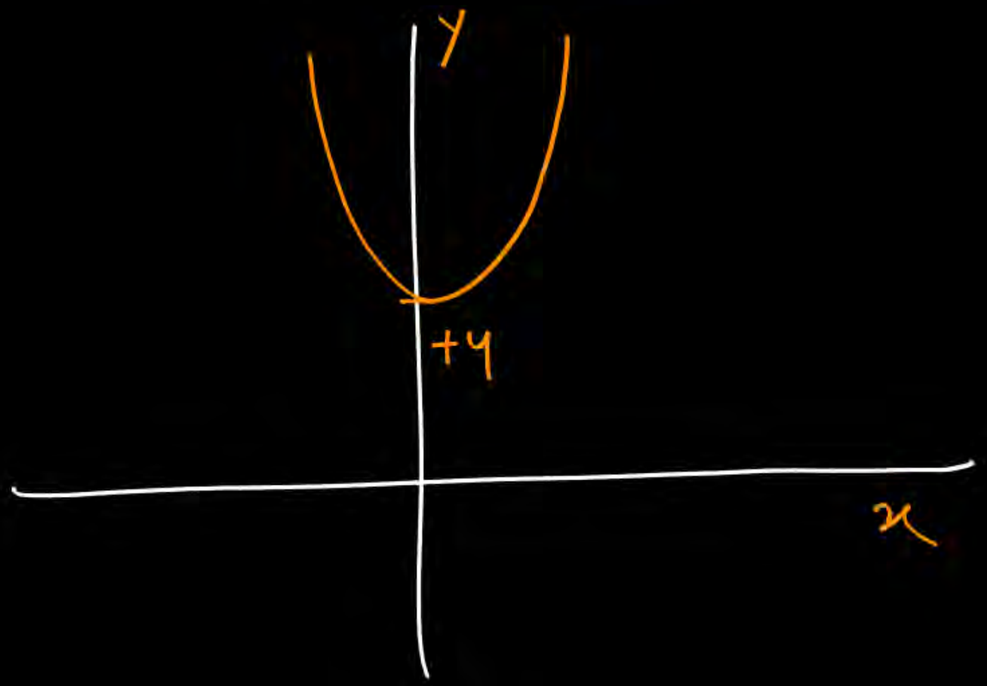
$$0 = x^2 - 4$$

$$4 = x^2$$

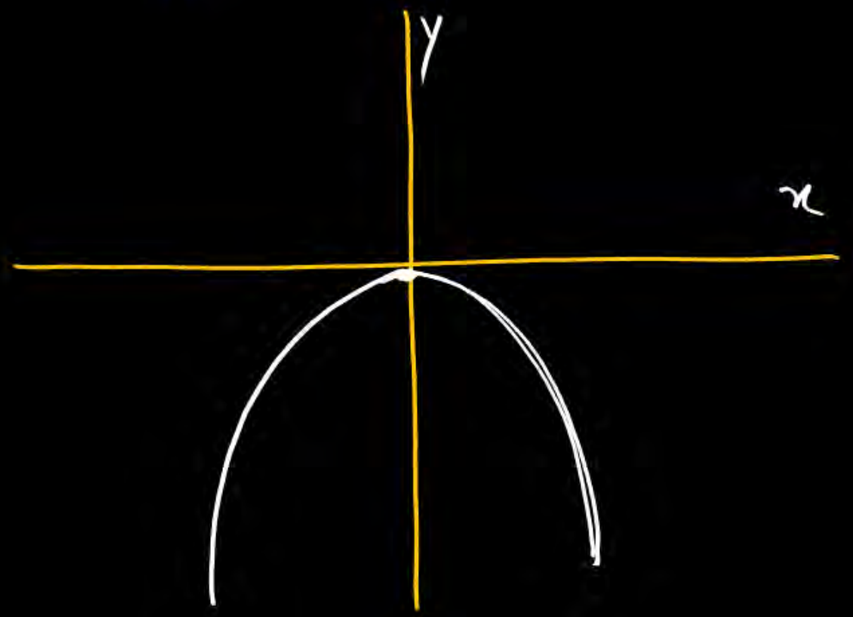
$$x = \sqrt{4}$$

$$x = \pm 2$$ ✓

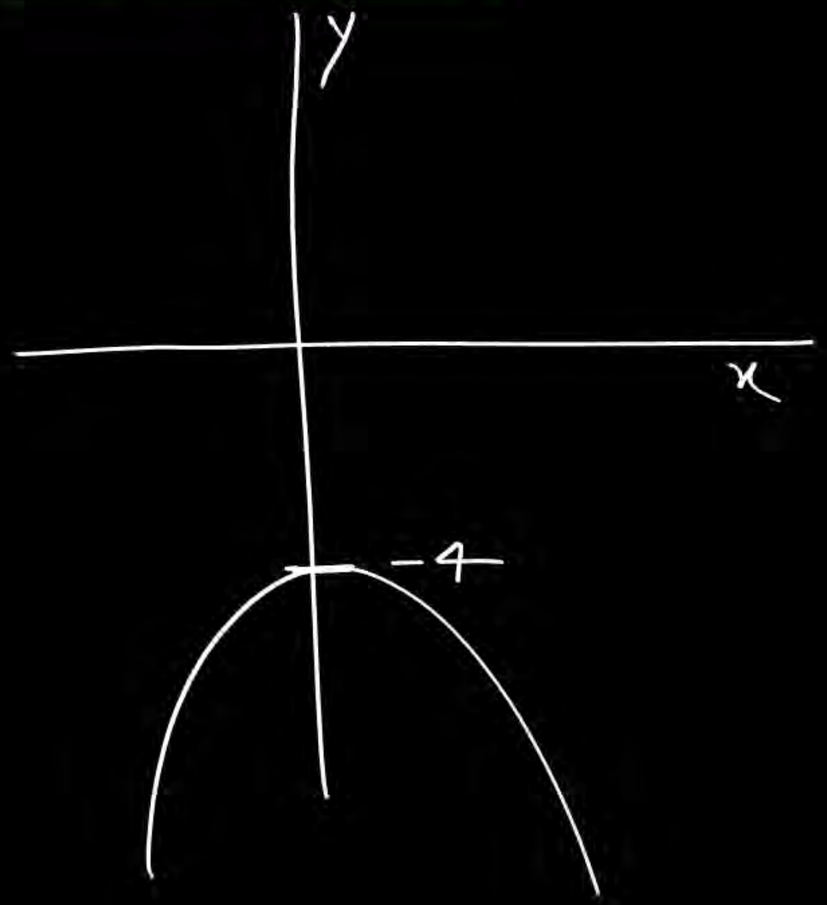
$y = x^2 + 4$



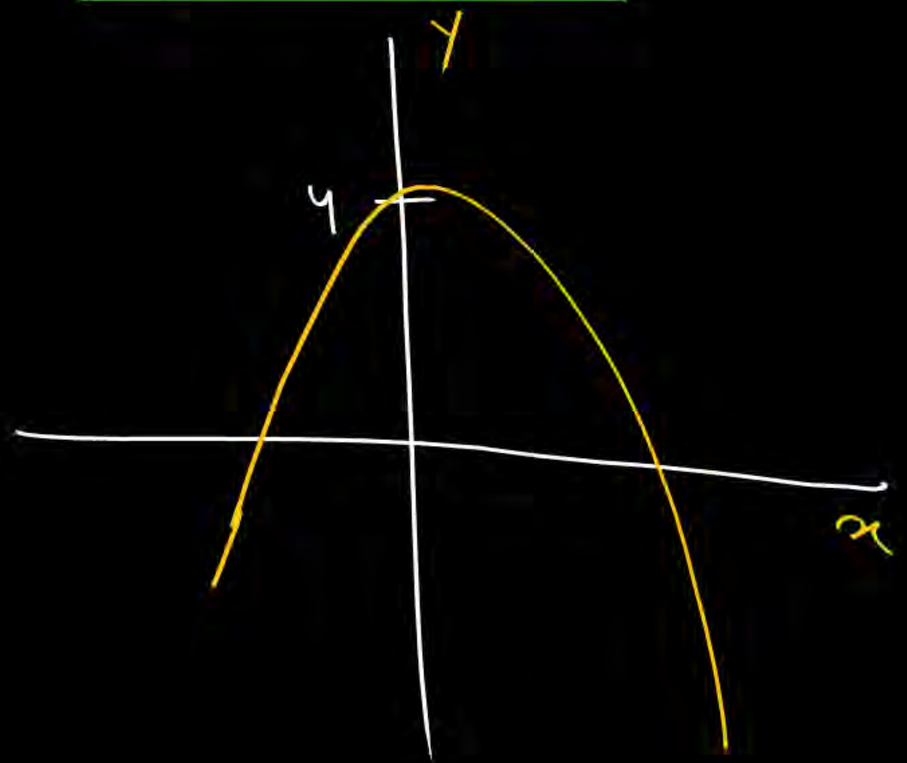
$y = -x^2$



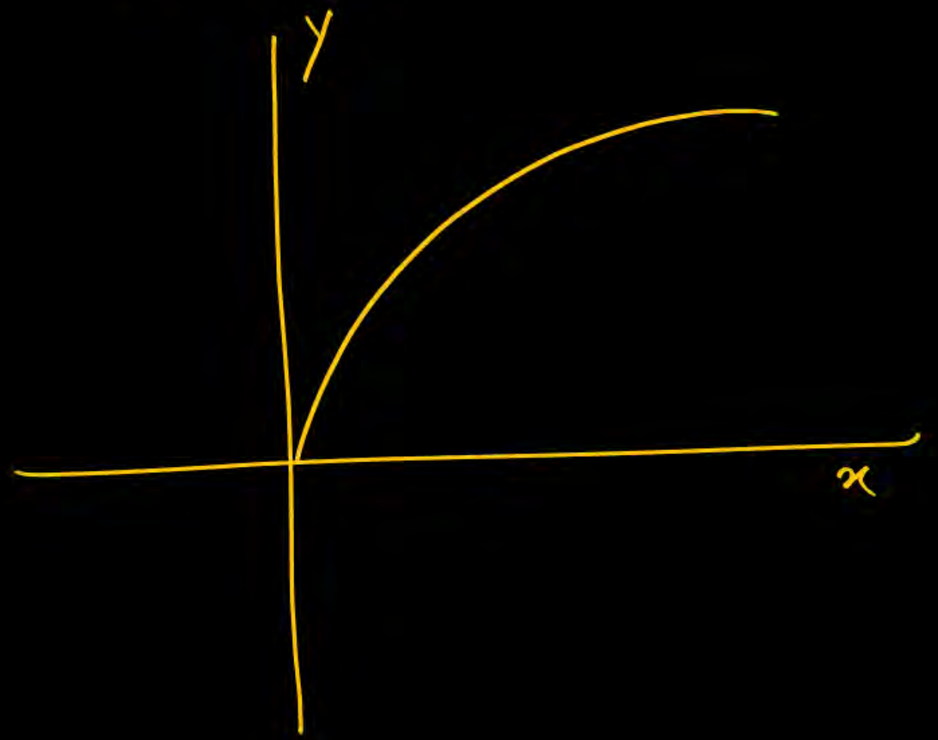
$$y = -x^2 - 4$$

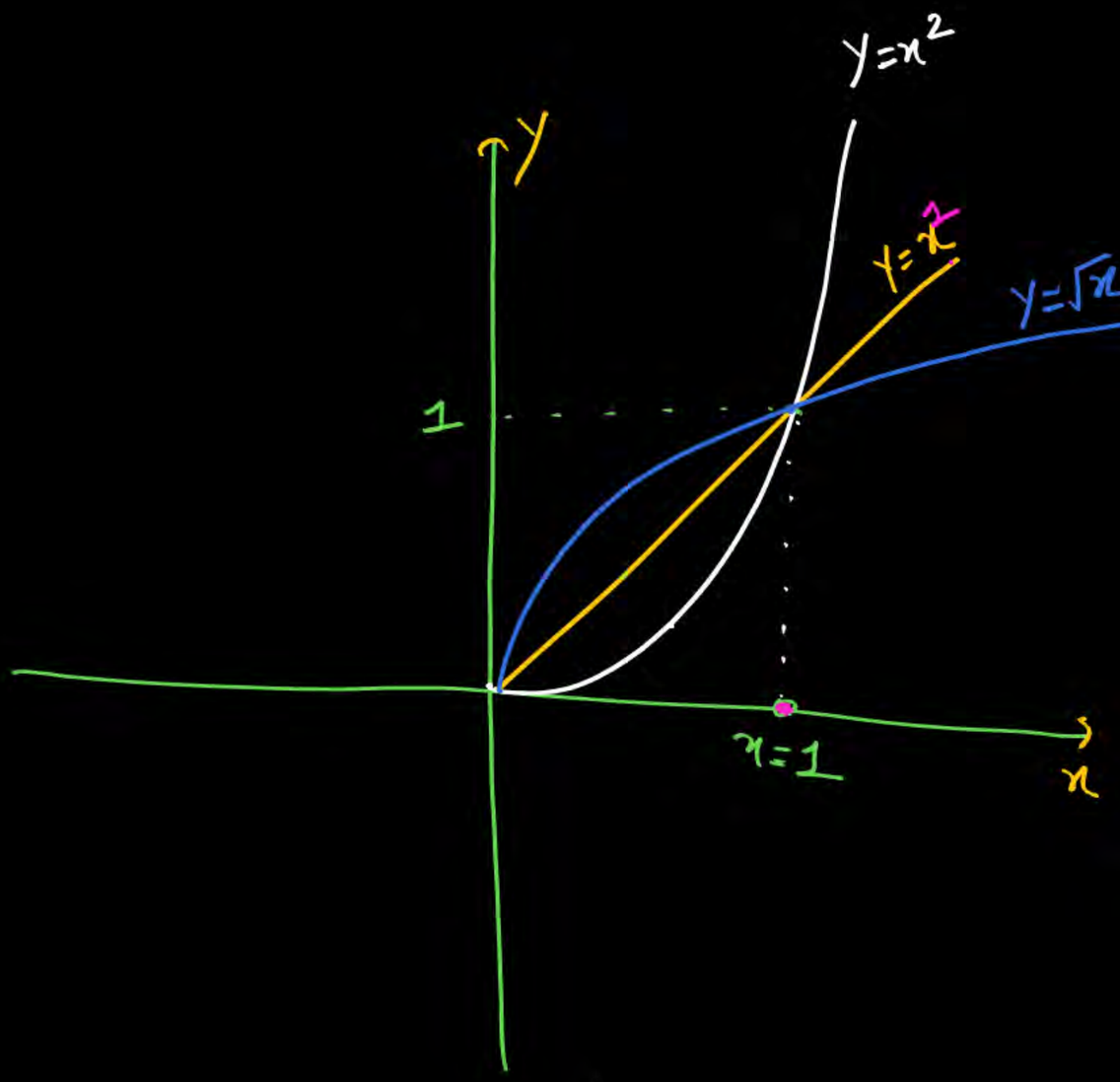


$$y = -x^2 + 4$$



$$y = \sqrt{x} = x^{1/2}$$



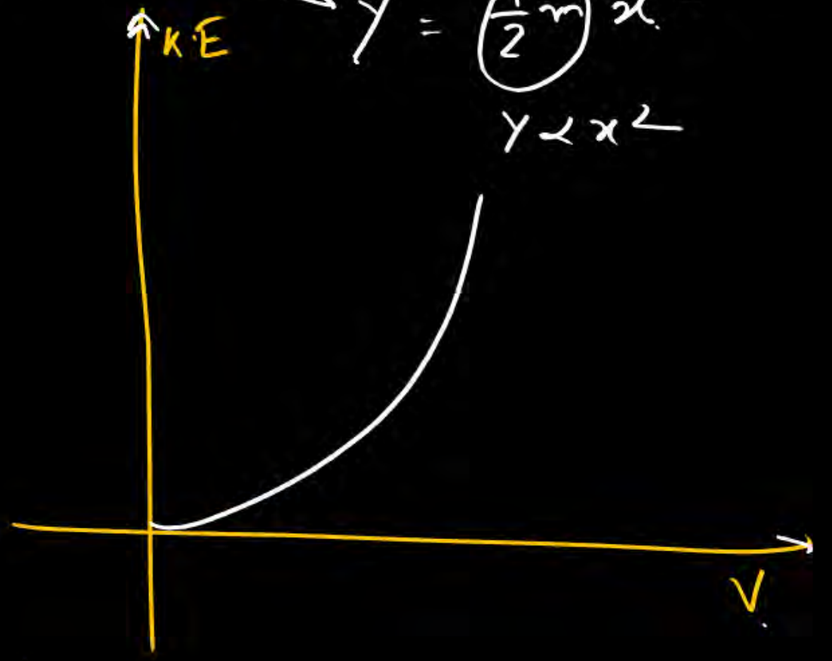


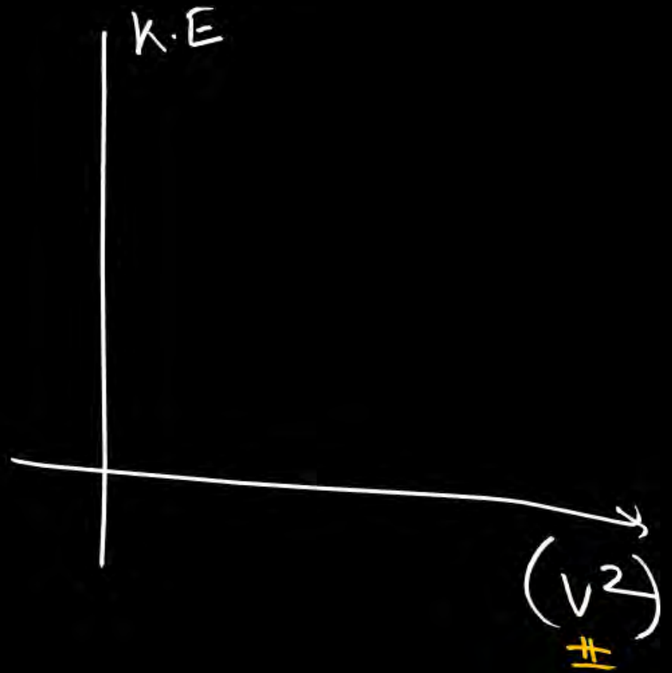
Draw graph b/w "K.E." and speed

$K.E = \frac{1}{2} m v^2$

$y = \left(\frac{1}{2} m\right) x^2$

$y \propto x^2$





Draw graph b/w Time & length of simple pendulum.

Keppler's law :- square of time period is propotional to cube of radius. $T^2 \propto R^3$. Draw graph b/w T & R.

If $K.E = \frac{p^2}{2m}$ Draw graph b/w K.E & mass

If $x^2+y^2=49$ Draw graph b/w x and y

Relation b/w velocity and Position of simple harmonic motion

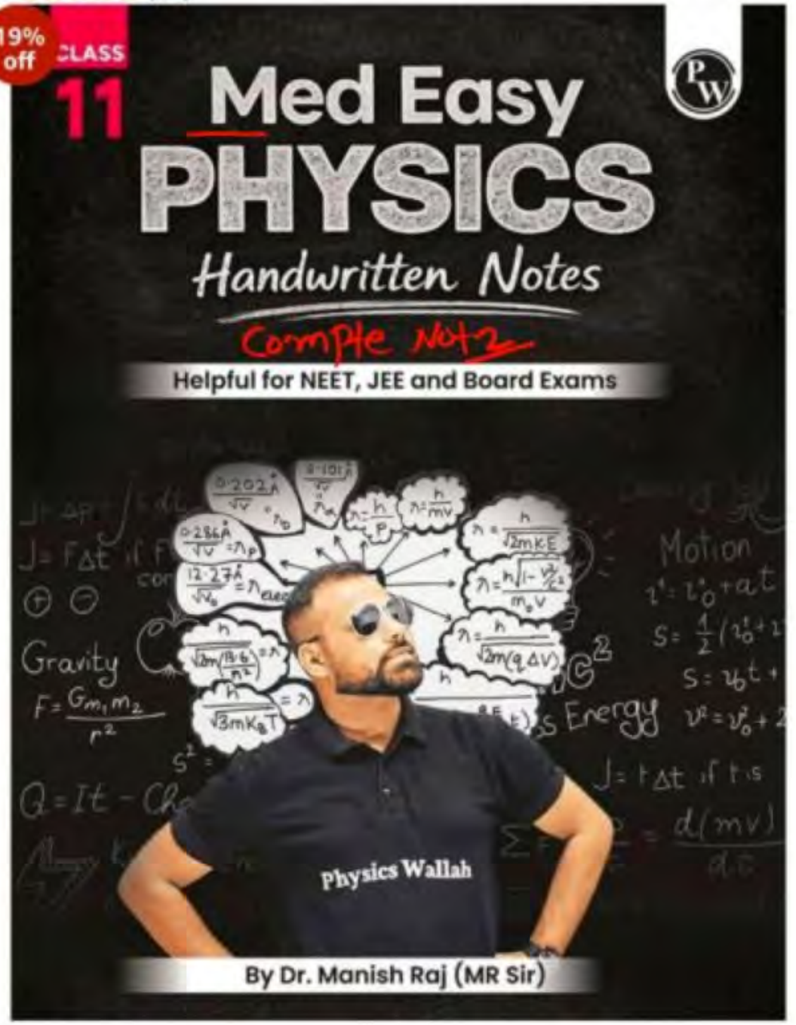
$$V = \omega \sqrt{A^2 - x^2}$$ where ω & A → cost^n then draw graph between v and x.



Sharsh Assignment 1 & 2

Join

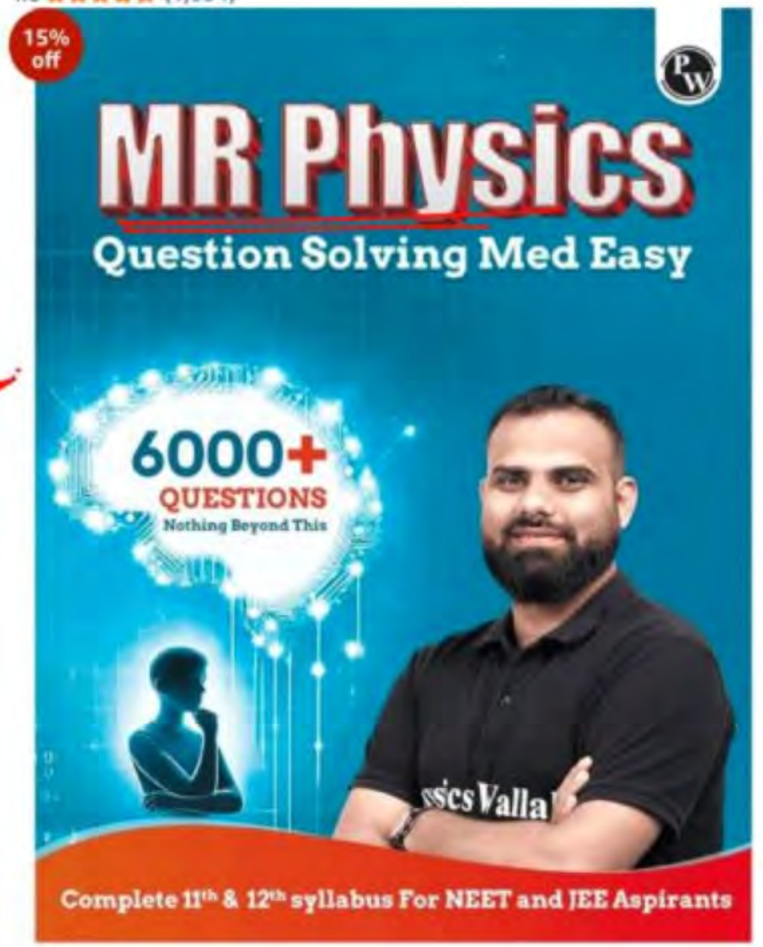
4.6 ★★★★½ (13)



5 July 2025

ISBN-13: 978-9371531382 ISBN-10: 937153138X

4.6 ★★★★½ (1,054)



Thank

You